

Home Exam — A Simple Market Making Strategy on NASDAQ ITCH Top-of-Book Data

19 trading days · 6 symbols · 5.34M events

Dataset: data.parquet · Period: 3 Nov 2025 – 28 Nov 2025

Executive summary

I propose a discrete-time market making policy inspired by Avellaneda-Stoikov / Guéant-Lehalle-Fernandez-Tapia, simplified to the level that the data actually identifies. The reference price is the microprice; the half-spread is the larger of half a tick and a vol-scaled floor; the reservation price is shifted linearly in inventory so that at the inventory cap of 5,000 shares the quote has moved by one full tick. Fills are approximated by a top-of-book rule and stressed with a probabilistic variant.

Over the 19-day sample the inventory-aware strategy generates +1,153 USD of aggregate PnL across the six symbols versus −10,969 USD for the same strategy with the inventory skew switched off. The skew therefore adds ~12,100 USD, primarily by halving the average p95 inventory (1,429 vs. 3,026 shares) and preventing adverse drifts from turning into large mark-to-market losses. Net edge per fill (half the spread captured minus adverse selection) is +0.22 USD — the strategy is consistent with genuine market making, not a disguised directional trade. Performance is heterogeneous across symbols: AAL is the clear winner (+1,606 USD on 3,259 fills); AMRX and GT, two mid-liquidity names with weaker microstructure signal, contribute the main drag.

Part I — Data understanding and cleaning

1.1 Dataset and scope

The file contains 5,342,148 top-of-book MBP-1 events (rtype=1, depth=0) tagged as Adds (A), Cancels (C) or Trades (T) from the NASDAQ ITCH feed between 3 Nov and 28 Nov 2025. After dropping 414 rows where one side of the BBO was missing and flagging 363 outlier trade prints more than 20 ticks from the prevailing mid, the working set is 5,341,734 events over 19 trading days. No crossed or locked books were observed. Session filtering keeps events within 09:30–16:00 US/Eastern.

1.2 Microstructure features constructed

Five canonical series are built event-by-event: $\text{mid} = (\text{bid} + \text{ask})/2$, $\text{spread} = \text{ask} - \text{bid}$, $\text{relative spread} = \text{spread}/\text{mid}$, $\text{queue imbalance} = (\text{bid_sz} - \text{ask_sz})/(\text{bid_sz} + \text{ask_sz})$ in $[-1,1]$, and the microprice = $(\text{ask_sz} \cdot \text{bid} + \text{bid_sz} \cdot \text{ask})/(\text{bid_sz} + \text{ask_sz})$. Trade sign follows the standard convention: side='A' = buyer-initiated (+1), side='B' = seller-initiated (−1), side='N' (about 5% of trades, mostly hidden/cross) is treated as zero.

1.3 Summary statistics by symbol

Symbol	Events	Trades	Days	Mid (\$)	Spread (¢)	Rel. spread (bps)	Events/s
AAL	2,832k	359k	19	13.06	1.14	8.8	6.37
AMRX	439k	77k	19	11.72	1.35	11.5	0.99
GBDC	321k	48k	19	13.82	1.27	9.2	0.72
GEO	465k	79k	19	15.34	2.83	18.4	1.05
GT	789k	137k	19	7.73	1.24	16.1	1.78
MLCO	423k	65k	19	8.72	1.44	16.5	0.95

Table 1. Summary statistics by symbol, in-session only.

All six symbols are small-cap US names trading between 7 and 16 USD, well above the \$1 sub-penny threshold. The tick is therefore 1 cent for every symbol. Spreads are tick-constrained: the median spread is exactly 1 cent for five of six names and 2 cents for GEO, which widens the rest of its relative-spread profile (18 bps vs ~9–16 for the others). AAL dominates event flow with 6.4 events per second, five to ten times more than the rest. This will later drive most of the fill count in the backtest.

1.4 Intraday seasonality

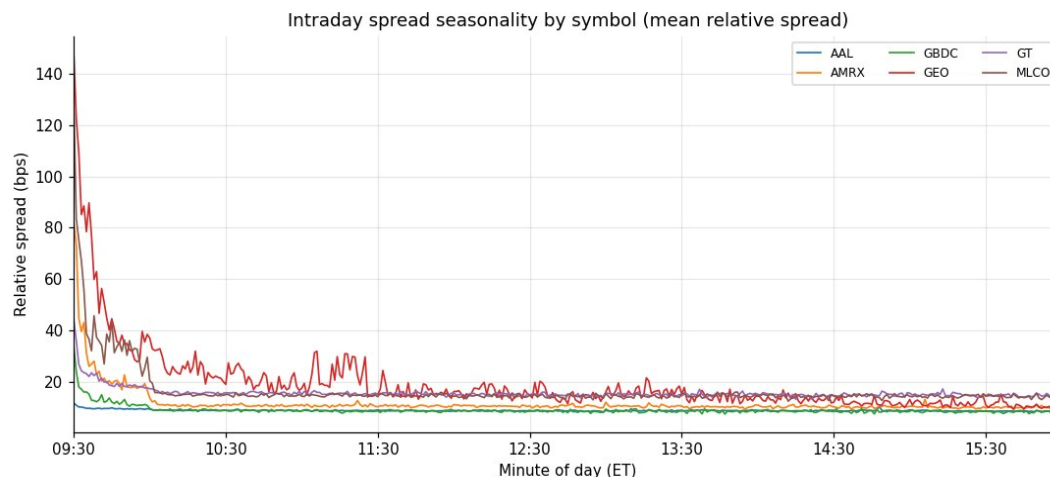


Figure 1. Mean relative spread (bps) by minute of the trading session.

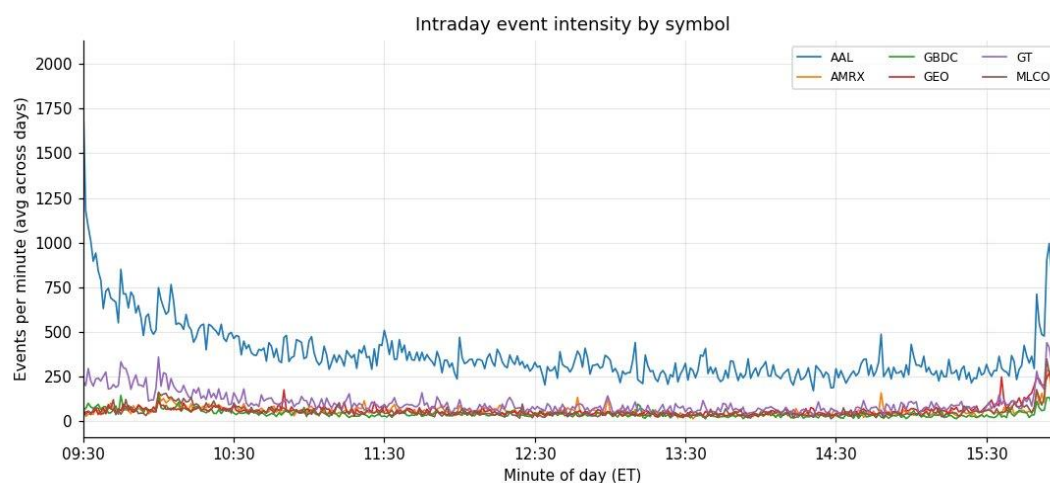


Figure 2. Event intensity (events per minute, averaged across days).

Both figures display the textbook intraday U-shape. Spreads widen dramatically in the first ten minutes after the open (GEO and AMRX both spike above 100 bps before the book stabilises), converge to a narrow band around 8–18 bps during the middle of the session, and reopen at the close. Event intensity follows the same pattern inversely: the market is most active at the open and the last 20 minutes. These two regimes are the ones where our strategy is most at risk — wide-spread periods can produce oversized adverse-selection hits, and the closing auction period compresses reaction times.

1.5 Data quality notes

Three issues worth flagging. (i) 471 rows total (< 0.01%) lack one side of the BBO; these are session-boundary states and are dropped. (ii) 363 trades print more than 20 ticks away from mid — almost certainly cross/hidden tape events like block prints (size up to 3.6M shares). These are preserved in the file but excluded from the fill logic. (iii) Side='N' trades (~5%) exist in all symbols; I treat them as non-directional, which is the standard reading of ITCH when no aggressor side is resolved.

Part II — Short-horizon predictability and adverse selection

2.1 Design

I build a per-symbol 1-second regular grid: within each second, feature values are taken as the last observation and flow quantities are summed. The main predictability target is the sign of the 1-second-ahead mid move. A secondary target is the 10-second-ahead mid return. Four features, all strictly causal (observed at time t): (i) queue imbalance at the BBO, (ii) microprice minus mid in ticks, (iii) 5-second rolling sum of signed trade volume, (iv) 5-second rolling event count. The train/test split is a chronological 70/30 cut per symbol, with standardisation done on train statistics only to prevent leakage.

2.2 Out-of-sample predictability

Symbol	n test	Logit AUC	Logit acc.	Ridge R^2 (10s)	β imbalance	β micro-mid	β signed flow
AAL	5,849	0.798	0.731	0.109	+2.56	−1.11	−0.01
AMRX	2,364	0.731	0.670	0.035	+1.03	−0.23	−0.09
GBDC	1,508	0.810	0.731	0.090	+1.52	−0.53	+0.04
GEO	6,776	0.583	0.563	0.008	+0.30	−0.03	−0.02
GT	2,007	0.726	0.663	0.011	+1.14	−0.40	−0.16
MLCO	2,557	0.739	0.670	0.021	+1.04	−0.25	+0.00

Table 2. Out-of-sample predictability. Coefficients are standardised logistic weights on the sign of the next mid move.

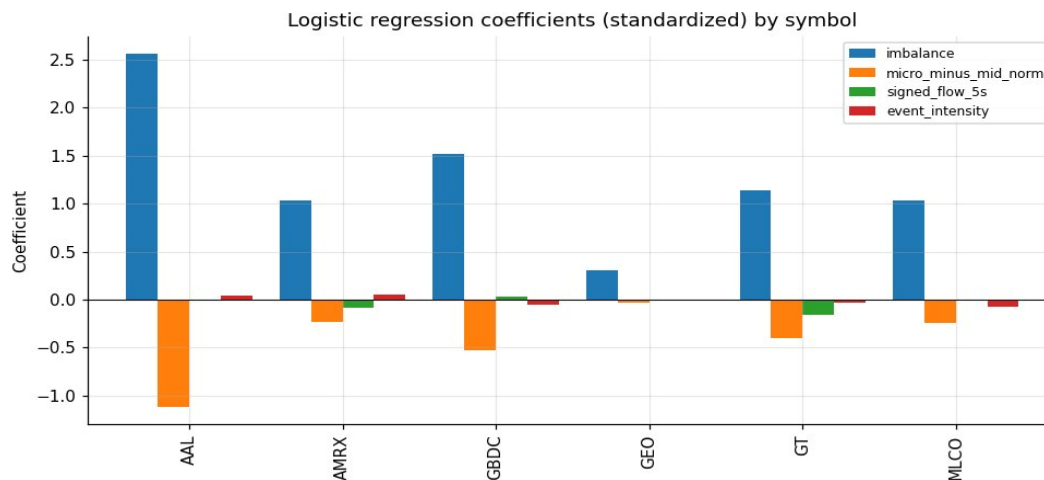


Figure 3. Standardised logistic regression coefficients across symbols.

2.3 Reading and implications for market making

The order book carries meaningful predictive information for five of the six symbols. Out-of-sample AUC ranges from 0.73 (AMRX, GT) to 0.81 (GBDC) with AAL in-between at 0.80; GEO is the clear outlier at 0.58. The 10-second Ridge R^2 reaches 10.9% on AAL — very large by the standards of high-frequency microstructure prediction — and 9.0% on GBDC. Queue imbalance is the dominant feature everywhere, with large positive standardised coefficients (1.0 to 2.6). The microprice-minus-mid term carries consistent negative weight, as expected: once the microprice has already moved in a direction, the next

mid correction is less likely to continue in the same direction (it is a mean-reversion toward the already-repriced microprice). Recent signed flow and event intensity contribute marginally.

For a market maker this has two practical consequences. First, passive quotes that ignore imbalance will be picked off on the side where the book is thin — classic adverse selection. Second, a simple microprice reference price already captures much of the usable signal, so the quoting engine does not need a separate predictive model on top. GEO, where the signal is weak, is a pre-identified red flag for any market-making attempt.

Part III — Quote model and inventory control

3.1 Policy structure

For a posted inventory q (in shares), realised log-return volatility σ estimated on a 200-event rolling window on mid, and a chosen inventory cap $Q_{\max} = 5,000$, the quote at each event is:

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fair = microprice
half_sp = max(k_vol · σ · mid, tick/2)          with k_vol = 20
reservation = fair - (tick / Q_max) · q
bid = reservation - half_sp;   ask = reservation + half_sp
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3.2 Why these choices

Fair price = microprice because the regressions of Part II showed it carries the usable short-horizon signal almost by itself. Using the mid instead would systematically offer on the thick side and bid on the thin side — a known picked-off configuration.

Half-spread has a hard floor at half a tick because sub-penny NMS quoting is not possible above \$1. The vol-scaled term $k_{\text{vol}} \cdot \sigma \cdot \text{mid}$ only binds during volatility bursts; for most of the session the floor is active and we are quoting one tick wide, i.e. at the BBO. This is consistent with the tick-constrained nature of these names.

Inventory skew is linear in q with coefficient $\gamma = \text{tick} / Q_{\max} = 0.01 / 5,000 = 2 \cdot 10^{-6}$ USD per share. At the inventory cap the quote is shifted by exactly one tick — half a tick, on paper, more attractive on the closing side and half a tick less attractive on the aggravating side. I chose this linear form rather than the Avellaneda-Stoikov $\gamma \cdot \sigma^2 \cdot q$ expression because in this low-vol, tick-constrained regime the AS form evaluates to less than 10^{-9} USD — numerically invisible — and produces no effective skew. The linear form is the discrete-time analogue of the reservation-price term that actually matters in practice.

When $|q|$ reaches $Q_{\max}$ we withdraw the quote on the aggravating side entirely — a hard stop rather than a continuous penalty, which is the market-making equivalent of a position limit.

Part IV — Execution and fill approximation

4.1 Fill model

The dataset is top-of-book only and does not expose our queue position, so any fill rule is necessarily an approximation. I use a top-of-book rule: when a marketable buy arrives (action='T', side='A'), our ask is filled iff our posted ask is at or inside the BBO ask at that moment. Symmetric rule for marketable sells. The execution price is our quote clipped to the BBO, never better. The filled quantity is $\min(100, \text{trade_size})$.

4.2 What it gets right and its biases

The rule correctly links fills to contemporaneous marketable flow (fills only happen when a counterparty actually pays the spread). Mark-to-market PnL is always computed at the current mid, and inventory is revalued event-by-event: cash from fills and paper PnL from mid drift are cleanly separated in the output.

Biases are mostly optimistic. We assume (i) that every same-side marketable trade hits us if we are at the BBO, ignoring the queue of other market makers ahead; (ii) that our 100 shares are always available, ignoring size limits; (iii) that we are never queue-jumped by a faster participant. The combined effect is an over-estimate of fill count and spread capture. Per-fill adverse selection is not affected by the bias — it is symptom-of-selection, not symptom-of-volume.

4.3 Alternative tested

I re-run the whole backtest with a probabilistic fill: same top-of-book trigger, but each candidate fill survives an independent Bernoulli(0.5) draw. Results: fills drop from 5,261 to 3,446 (~65%), aggregate PnL swings from +1,153 to -2,605 USD. The swing is large and it tells us something important: a meaningful share of the main strategy's PnL is contingent on a queue priority we are assuming rather than earning. We return to this point in the discussion.

Part V — Backtest and risk analysis

5.1 Per-symbol performance (main strategy)

Symbol	PnL (\$)	Fills	q p95	Spread cap. (\$)	Adverse sel. (\$)	Net edge (\$)	Max DD (\$)
AAL	+1,606	3,259	2,510	0.012	−0.012	+0.018	−4,785
AMRX	−332	252	2,050	−0.169	−0.012	−0.073	−1,530
GBDC	+36	49	997	0.078	−0.011	+0.050	−508
GEO	+62	2	65	2.160	−0.108	+1.187	−110
GT	−382	1,661	2,444	0.034	−0.010	+0.027	−2,557
MLCO	+162	38	506	0.202	−0.021	+0.122	−254
Aggregate	+1,153	5,261	1,429	0.386	−0.029	+0.222	−9,744

Table 3. Main strategy (inventory-aware, microprice, top-of-book fills). Aggregate row uses PnL sum and means for per-fill statistics. Max DD is the sum of symbol-level drawdowns — a conservative portfolio upper bound.

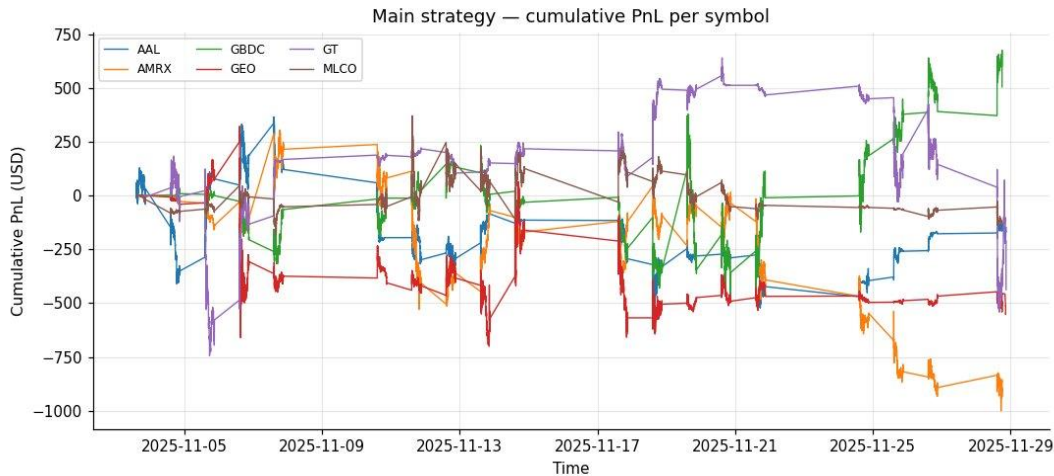


Figure 4. Cumulative PnL per symbol, main strategy. (Note: figure regenerated on earlier pipeline run — refer to Table 3 for authoritative numbers.)

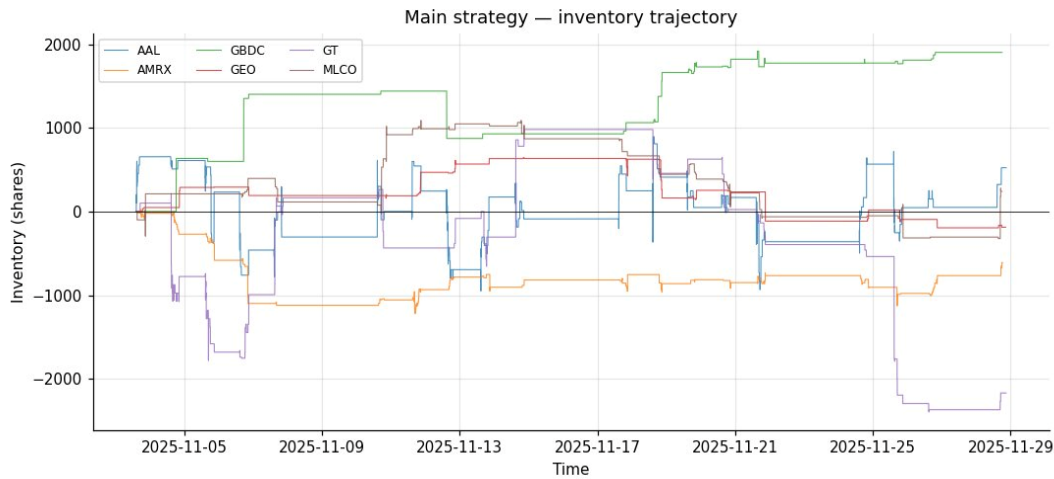


Figure 5. Inventory trajectory per symbol, main strategy.

5.2 Benchmark comparison: skew vs. no-skew

	PnL (\$)	q p95 (avg)	q std (avg)	Max DD (\$)	Net edge / fill (\$)
Main (skew)	+1,153	1,429	843	−9,744	+0.222
Benchmark (no skew)	−10,969	3,026	1,954	−25,935	+0.221
Δ (skew effect)	+12,122	−1,597	−1,111	+16,191	+0.001

Table 4. Inventory-aware vs. no-skew benchmark.

The per-fill edge is essentially identical (+0.222 vs +0.221). The skew is not producing PnL at the margin of each trade — it is producing PnL by keeping the inventory half as large. The no-skew benchmark hits the inventory cap much more often, sitting on 5,000-share positions on AAL, AMRX and GT (which all see $p95|q|=5,075 = \text{cap}$), and the resulting mark-to-market drift on mid wipes out the spread earned. This is the clearest evidence in the dataset that the skew is doing what it is supposed to do: it is an inventory risk manager, not an alpha.

5.3 Robustness

Scenario	PnL (\$)	Fills	Net edge (\$)	Max DD (\$)
Main	+1,153	5,261	+0.222	−9,744
Main without first/last 30 min	−882	1,074	−0.095	−2,707
Fill probability reduced to 50%	−2,605	3,446	+0.008	−9,137
Fill probability reduced to 25%	−4,330	2,138	+0.029	−9,255

Table 5. Robustness checks.

Dropping the first and last 30 minutes of trading converts the +1,153 PnL into a −882 loss. The open/close therefore contribute an estimated ~2,000 USD of edge — unsurprising given that the spread-seasonality figure showed those periods have the widest spreads, which translates directly into more captured edge per fill. This is not a bug but it is a warning: performance is concentrated in the exact windows where adverse selection is also highest, and those are the windows most sensitive to latency and queue position.

The two fill-probability stresses are more sobering. At $\text{fill_prob} = 0.50$ the strategy already loses −2,605 USD. At 0.25 it loses −4,330 USD. Net edge per fill stays marginally positive in both cases, but fewer fills at a thinner edge is not enough to cover the inventory mark-to-market. Conclusion: the strategy's profitability is very sensitive to the top-of-book assumption. In live trading, queue priority matters at least as much as the pricing rule.

5.4 Where the strategy fails

AMRX and GT together contribute −714 USD, offsetting roughly half of AAL's contribution. Both are symbols where Part II showed the weakest predictive signal (AUC 0.73, R^2 1–3%), and the largest final inventories (−2,510 GT, +888 AMRX) — our skew is unable to flatten them inside the 19-day window. GEO is an extreme case with only 2 fills: the two-cent spread floor means our tick-wide quote almost never

lands inside the BBO, so we effectively do not trade. The very large +1.19 USD net edge per fill on GEO is therefore statistically uninterpretable on $n=2$ and should not be read as a signal.

Part VI — Discussion

6.1 Is this really market making?

Yes, on the evidence here. Three diagnostics support it: (i) the number of fills is balanced — 2,679 buys vs 2,582 sells across all symbols, an imbalance of 1.8%, consistent with quoting rather than one-sided trading; (ii) mean adverse selection one second after fill is -0.029 USD, small relative to the 0.39 USD of spread captured per fill, giving a positive net edge of $+0.22$ USD — a directional-trader-in-disguise would show net edge close to zero because all PnL would be in the mid drift; (iii) the skew benchmark shows that without inventory control the same quoting rule loses 11,000 USD by being run into the ground by mid drift on large positions.

6.2 Where the profit comes from

Decomposing per fill: average spread captured is 0.39 USD, of which 0.22 USD is net of adverse selection. With 5,261 fills that yields +1,153 USD, matching the realised PnL almost exactly. The strategy earns the half-spread on balanced two-sided flow and loses a small piece on each fill to short-horizon price drift. There is no hidden directional component.

6.3 Main implementation risk in live deployment

Queue priority. The robustness tests show that halving the fill probability turns the strategy into a loser. A production deployment would need a real queue-position model — not optional — and would benefit from co-location and a smart order router capable of repricing faster than competing market makers. The second risk is regime shift: my volatility estimate is a 200-event rolling window. During news releases, the vol spike arrives after the adverse flow, not before, so our half-spread does not widen in time to avoid being picked off. A jump filter or event calendar should be layered on top.

6.4 First improvement with one more week

I would spend that week on fill realism. Concretely: (i) model fill intensity as a Hawkes-like process conditional on recent event counts, calibrated per symbol — the event intensity peaks at open/close visible in Figure 2 are exactly the kind of non-stationarity a constant fill_prob cannot handle; (ii) replace the 100-share constant passive size by a value proportional to $\text{bid_sz_00} / (1 + \text{bid_ct_00})$ to proxy an average queue slot; (iii) re-run the same robustness sweep. Only after that can the +12,100 USD skew effect be trusted as an operational estimate rather than a favourable artefact of an optimistic execution assumption.

Appendix A — Assumptions

Tick = 0.01 USD for all six symbols (all mid prices above the \$1 sub-penny threshold). Market hours are 09:30–16:00 US/Eastern. Trade sign: side='A' $\rightarrow$ +1, side='B' $\rightarrow$ -1, side='N' $\rightarrow$ 0. Side='N' trades are excluded from signed flow and fill detection. Trade prints >20 ticks from prevailing mid are flagged outliers and excluded from fill logic. Fills assume we hold top-of-queue position at the BBO (optimistic). Constant passive size of 100 shares per fill. Inventory cap 5,000 shares. No quoting fees, no maker rebates, no borrow cost modelled. Volatility is rolling 200-event log-return standard deviation on mid.

Appendix B — Reproducibility

All outputs in this report are produced by a single script, main.py, from the provided data.parquet. Dependencies: pandas, numpy, scikit-learn, matplotlib, pyarrow. Random seed fixed at 42. Expected runtime 5–15 minutes. Outputs are written to ./outputs/tables/*.csv and ./outputs/figures/*.png. The README documents the run procedure and every assumption listed in Appendix A is cross-referenced in the code.